Computer Vision and Deep Learning

Lab 05: Semantic segmentation

Semantic segmentation

- ***Semantic segmentation*** is a technique that enables us to associate each pixel of an image with a class label, such as trees, cars, sky, etc

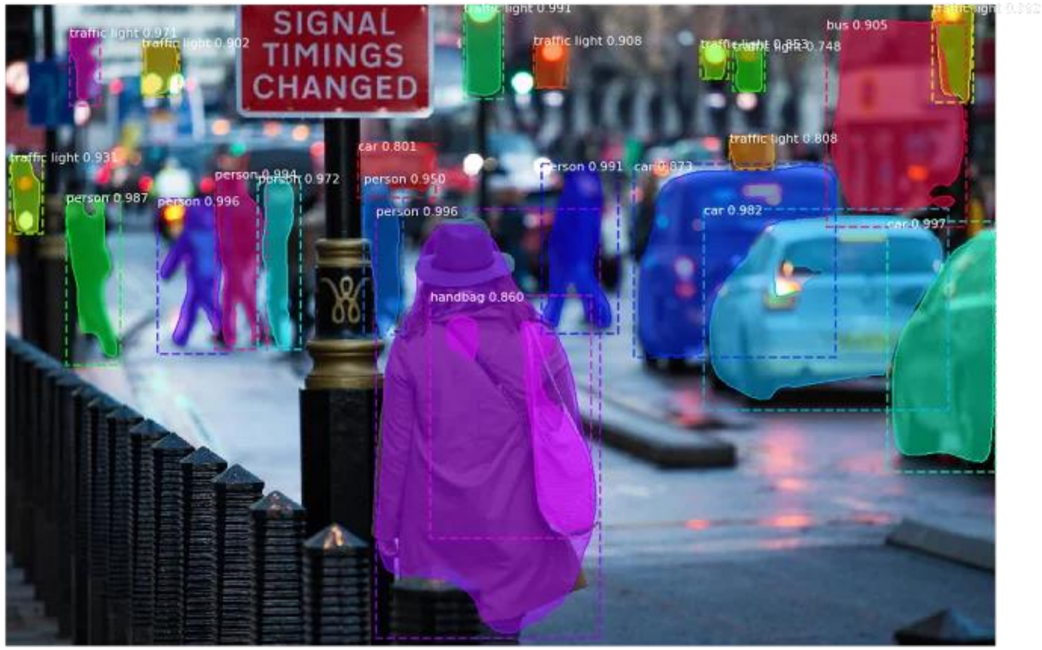
- It is also considered an image classification task at a pixel level as it involves differentiating between objects in an image

- It is essential to understand that semantic segmentation ***classifies image pixels of one or more classes without giving a semantically meaning at each object***

Instance segmentation

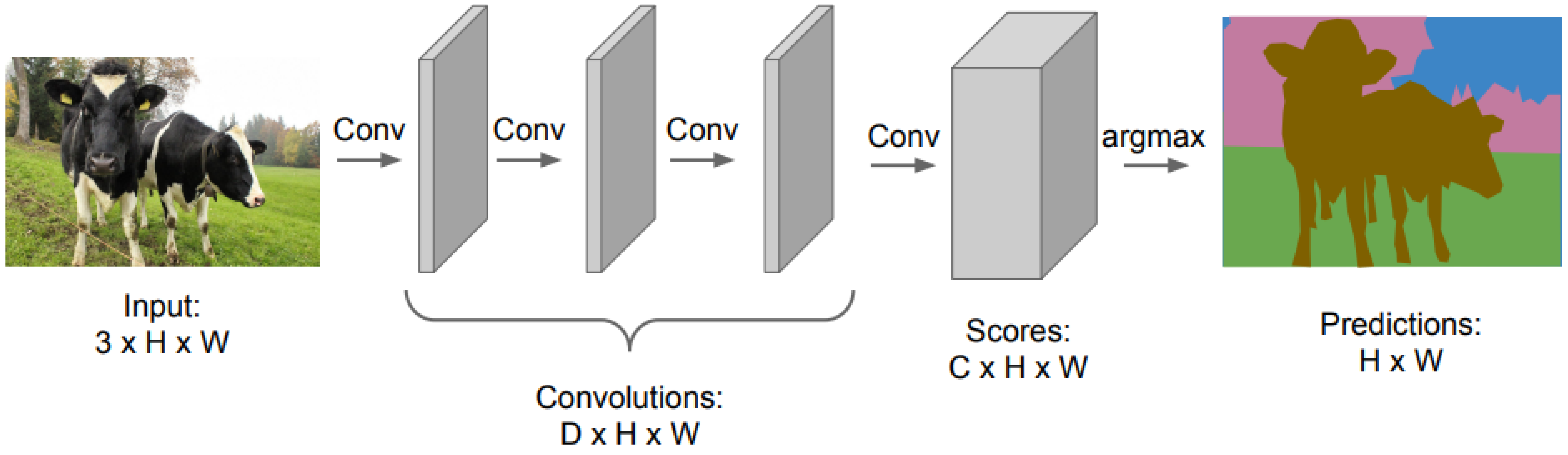
- ***Instance Segmentation*** is a unique form of image segmentation that deals with detecting and delineating each distinct instance of an object appearing in an image

- Instance segmentation ***detects all instances of a class with the extra functionality of demarcating separate instances of any segment class***



Semantic segmentation: fully convolutional

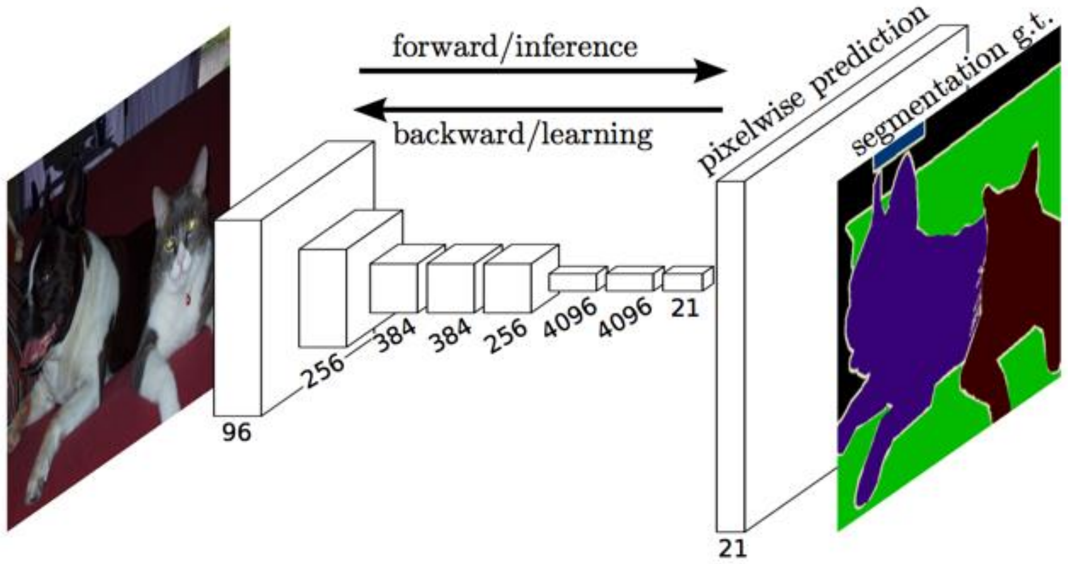
- A ***naive approach*** towards is to ***simply stack a number of convolutional layers*** (with same padding to preserve dimensions) and ***output a final segmentation map***



[*] https://arxiv.org/abs/1411.4038?ref=jeremyjordan.me

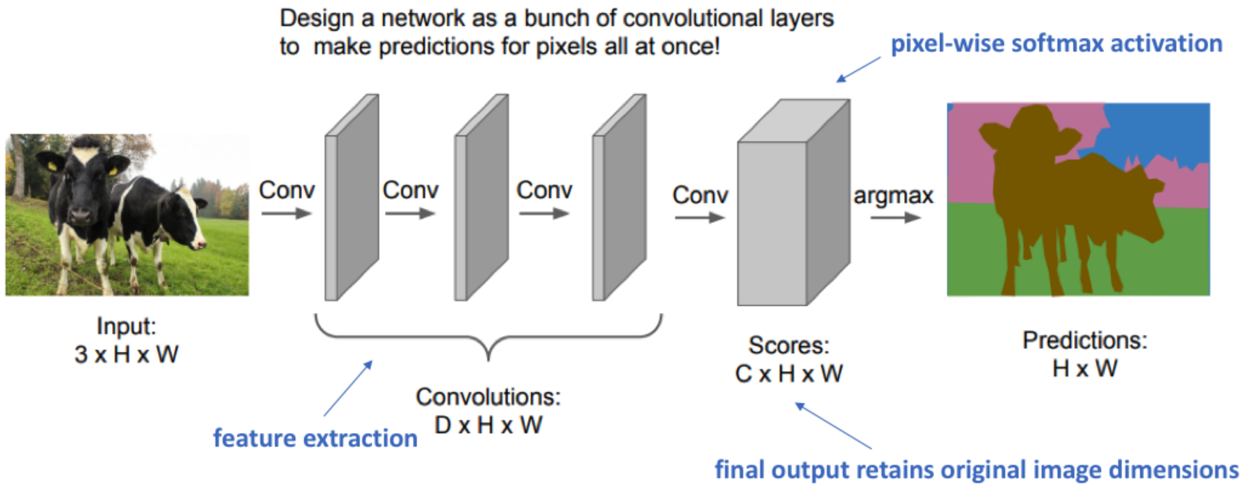
Semantic segmentation: fully convolutional

- ***The full network***, as shown below, ***is trained according to a pixel-wise cross entropy loss***



Semantic segmentation: fully convolutional (cont'd)

- This directly learns a mapping from the input image to its corresponding segmentation; however, ***it's quite computationally expensive to preserve the full resolution throughout the network***



Downside: Preserving image dimensions throughout entire network will be computationally expensive.

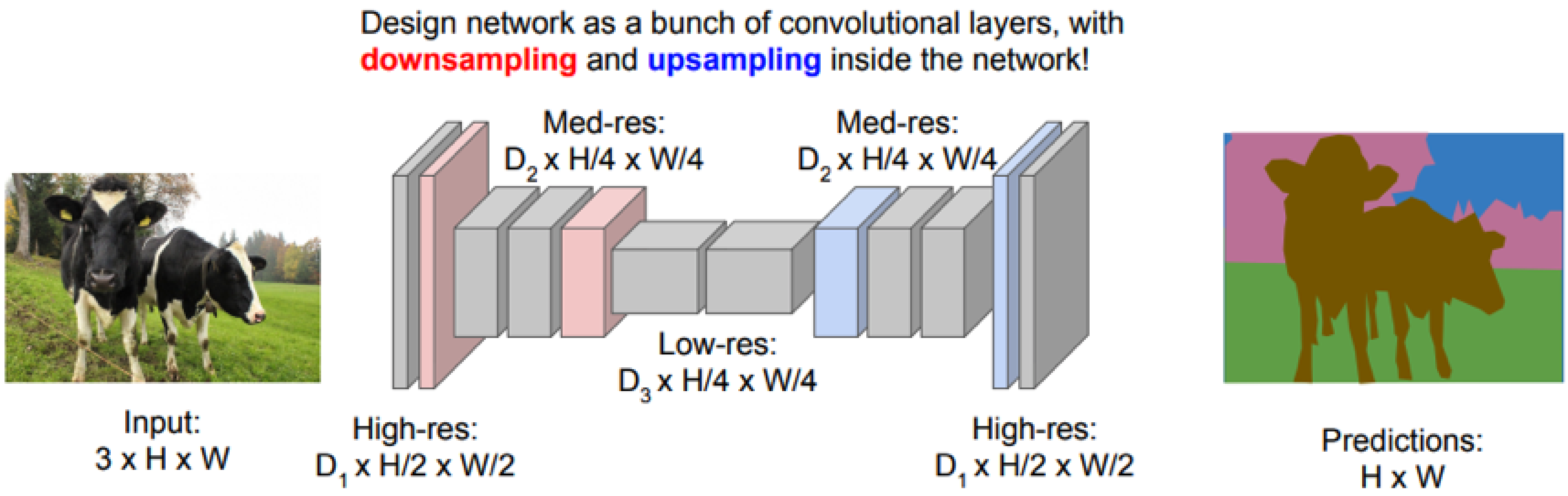
Semantic segmentation: encoder/decoder

- One popular approach is to use an ***encoder/decoder structure***

- Here we ***downsample*** the spatial resolution of the input, developing lower-resolution feature mappings which are learned to be highly efficient at discriminating between classes

- And then, **upsample** the feature representations into a full-resolution segmentation map

Semantic segmentation: encoder/decoder (cont'd)



Solution: Make network deep and *work at a lower spatial resolution* for many of the layers.

Skip connections

- Because deep neural networks can “forget” certain features as it pass information through successive layers, skip connections can reintroduce them to make learning stronger

- Skip connection was introduced in Residual Network (ResNet) and showed classification improvements as well as smoother learning gradients

The U-Net architecture

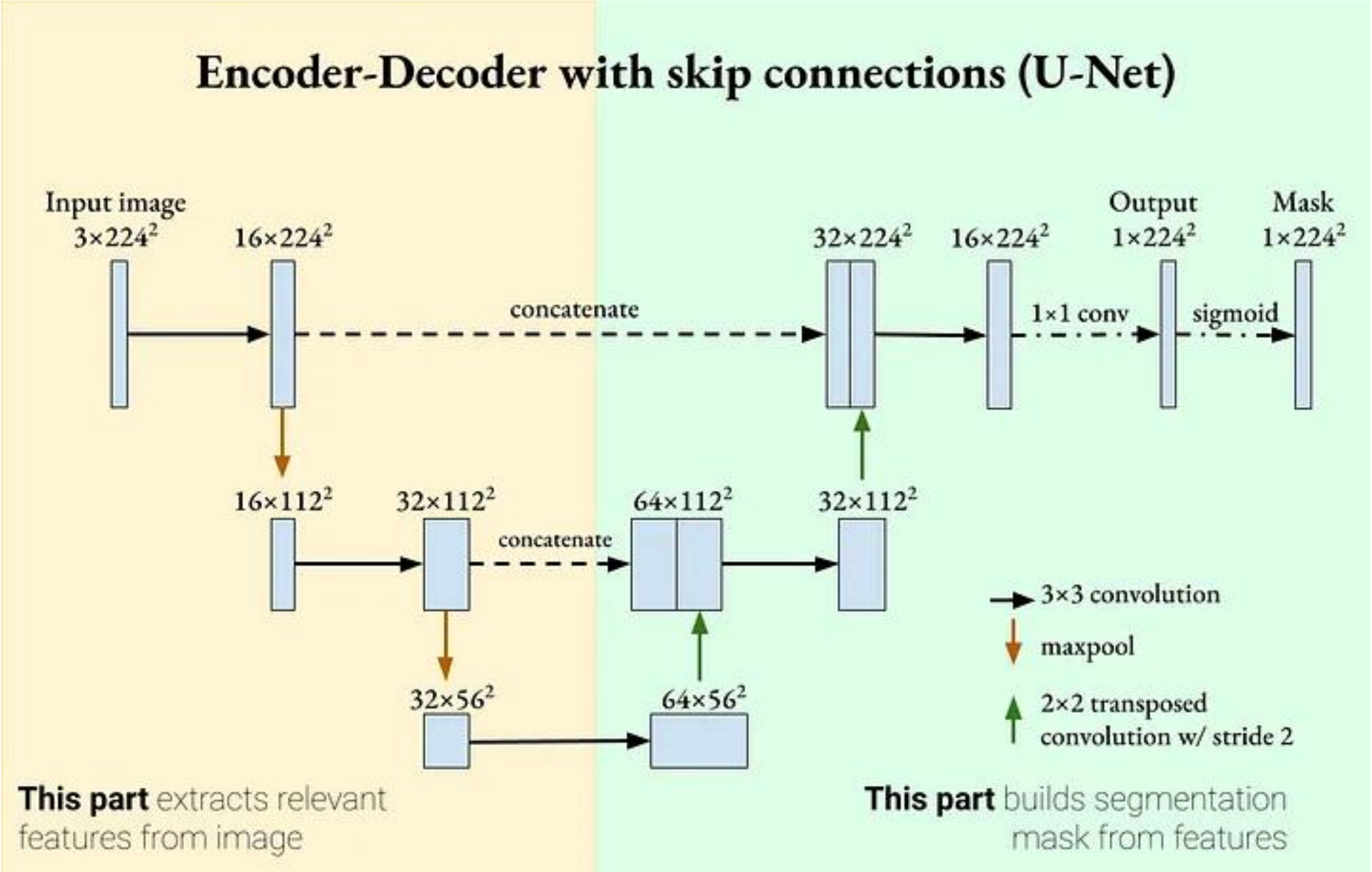
- Ronneberger *et al.* [1] expanding the capacity of the decoder module of the network

- More concretely, they propose the U-Net architecture which *"consists of a contracting path to capture context and a symmetric expanding path that enables precise localization."*

- This simpler architecture has grown to be very popular and has been adapted for a variety of segmentation problems

[1] https://arxiv.org/abs/1505.04597?ref=jeremyjordan.me

The U-Net architecture



Exercises